Satrio Adi Prabowo

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Summary

Senior Data Scientist with 7+ years of experience building production-grade ML systems across fraud detection, AML, and recommendation systems. Hands-on across the full stack from graph feature engineering to agentic AI systems replacing manual workflows. Consistent track record of delivering rigorous, end-to-end solutions with clean, maintainable code.

Experience

Data Scientist August 2026 – Present

TyrAds *Bali, Indonesia*

- Owned the fraud prevention system for TyrAds' rewarded ad network. Recalibrated the production fraud score, reducing block rate from 3.91% to 2.74% (−1.17 pp, −30% relative) while holding downstream conversion rejections (anomalous level progression, purchase fraud) at 0.44%.

Senior Data Scientist May 2022 – August 2026

DANA Indonesia (PT. Aliansi Teknologi Indonesia) *Bandung, Indonesia*

- Built AI Cyber Patrol agent that replaced 15 human agents, autonomously scanning 2,100 potential sites/month to detect DANA account numbers and QRIS merchants involved in illegal online gambling.
- Built AML alert prioritization using scorecard (merchant) and CatBoost (user) models enriched with PEP/sanction and gambler signals via tabular and Cypher graph queries, reducing investigation subjects by 98% (30K → 500) per cycle.
- Initialized and deployed a fraud/scam score model; enhanced it by 6% through graph feature integration.
- Developed trusted location system profiling users' frequent payment coordinates to detect geographic anomalies and prevent account takeover (ATO).
- Developed one-to-many facial recognition for duplicate selfie detection during KYC verification; trained on 70K face triplets, achieving 7% Equal Error Rate (EER).
- Rewrote existing recommendation system in PySpark, achieving 12x reduction in processing time.
- Enhanced recommendation system with Learning-to-Rank, improving NDCG by 4% for 80M+ active users.
- Developed personalized DANA service suggestions for 80M+ active users, enhancing overall user experience.
- Led data science collaboration workflow improvements, establishing best practices in code structure and project management.

Data Scientist Sep. 2021 – May 2022

Dagangan (PT. Dagangan Karya Indonesia) *Remote (Yogyakarta)*

- Developed a KTP and selfie visibility model to support paylater service initiation.
- Utilized AWS Textract OCR for data extraction from KTP photos, improving data processing efficiency.
- Implemented a Traveling Salesman Problem solver to optimize driver and sales route suggestion.
- Supervised Data Science Interns in building a recommendation system.

Data Scientist Sep. 2020 – Sep. 2021

eFishery (PT. Multidaya Teknologi Nusantara) *Remote (Bandung)*

- Evaluated credit scoring system, found weaknesses on credit score development process and recommended a better credit scoring guideline.
- Built propensity to transact model and propensity to churn model to enhance Sales Team's Roadmap Planning Process (RMP).
- Built a budget forecasting using SARIMA to forecast next 7-days budget for a better financial planning on a particular business unit.
- Built a farmer segmentation using k-means and extracted farmer cluster behavior for a better market targeting and understanding.

Data Scientist Jun. 2019 – Sep. 2020

Kredivo (PT. Finaccel Teknologi Indonesia) *Jakarta, Indonesia*

- Developed address passive-verification using k-means to cluster users' geospatial history data. This product increased 5% daily user acquisition.

- Developed user address prioritization, a pooling and ranking of bad users' valid address for debt collection purpose.
- Developed referral code social network analysis. The network can find a hidden potential abuse and fraud on community network level.
- Improved and enriched user income prediction performance by using XGBoost Regressor.

Data Scientist Mar. 2018 – Apr. 2019

Kumparan (PT. Dynamo Media Network) *Remote (Jakarta)*

- Established an article scoring system as the foundation for Kumparan's recommendation system, ensuring quality content pooling.
- Developed a personalized article feed based on content similarity using an unsupervised nearest neighbor algorithm.
- Developed A/B test analysis dashboard with Dash Plotly, enabling the Data and Product Teams to oversee ongoing experiments with ease.

Technical Skills

Tools: Python, Scikit-Learn, PyTorch, PySpark, Pandas, SQL, git, and exposure to GCP/AWS/Alibaba Cloud Environment

Domains: Machine Learning, Data Science, Deep Learning, Recommender System, Fraud Detection, AML, Graph Data Science, Credit Scoring

Certification

CompTIA DataX | *DY0-001* 2025

- Verify: verify.CompTIA.org — b33553cc67d54ad2b3a6a1803dcd7bed

Education

Telkom University, School of Computing Bandung, Indonesia

Bachelor of Informatics Engineering — GPA: 3.74/4.00 *2014 – 2018*

- Undergraduate Thesis: An Implementation of Support Vector Machine on the Multi-Label Classification of English-Translated Quranic Verses.
- Active AI/ML Teaching Assistant, Fundamental of Programming and Data Structure Practicum Assistant.
- Contributed to Cryptography Research Group of Computing Laboratory.